

Analysis summary

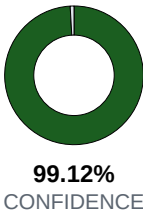
AI model: VGG16 (transfer learning, 4-class head)
Input: Single 2D slice, resized to 128×128 RGB
Pipeline: Inference → occlusion XAI → LIME overlay
Classes: Glioma · Meningioma · Pituitary · No Tumor

Diagnosis

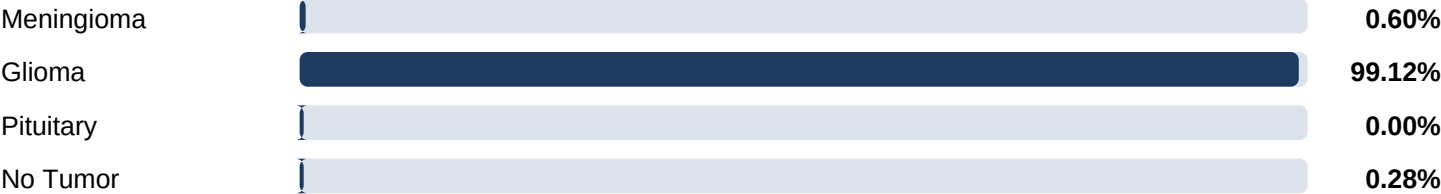
PREDICTION

Glioma Detected

Raw label: Tumor: glioma
Top-class confidence: 99.12%

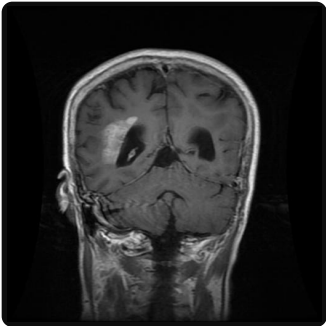


Class probabilities



Visualizations

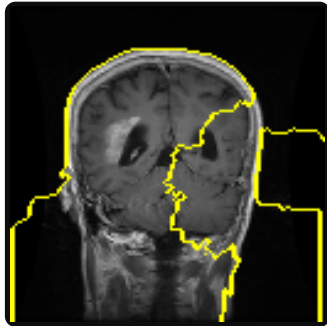
Original MRI



XAI heatmap (model attention)



LIME explanation



Findings & analysis

Model interpretation

Gliomas arise from glial cells and can appear as heterogeneous masses with variable intensity on MRI. The model's signal aligns with patterns commonly associated with this class in the training distribution.

LIME explainability summary

LIME perturbs superpixels and fits a local linear model to approximate the classifier near this image. Boundaries mark regions that pushed the score toward the predicted class (Glioma). Use alongside clinical context.

Report notes

- For research, education, or demonstration only.
- Not FDA-cleared; not a substitute for a radiologist.
- Single-slice uploads omit 3D context and sequence.
- Verify against the full MRI study and patient history.

Technical details

Architecture: VGG16 backbone + dense classifier
Input resolution: 128 × 128 (preprocessed)
Explainability: Occlusion sensitivity map; LIME
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